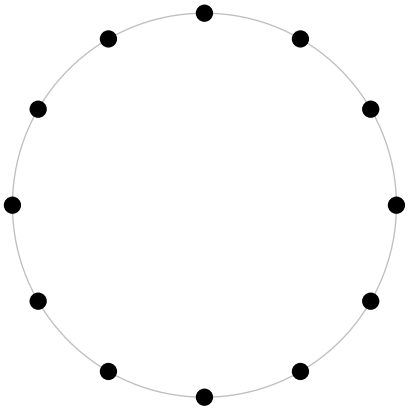
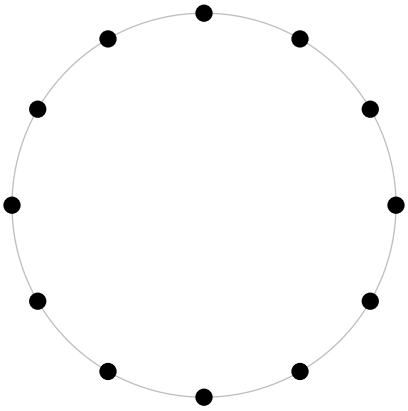
To draw hop 3 on a ring of dots, start at any dot and draw a straight line to the dot 3 places clockwise. Keep hopping 3 until you are back where you started. If some dots still have no line, start again at one of them, and stop when every dot has a line. The lines you draw from one start make one piece.

Problem 1: Draw hops 1 to 6 on these rings of 12 dots. Write how many pieces each drawing has.



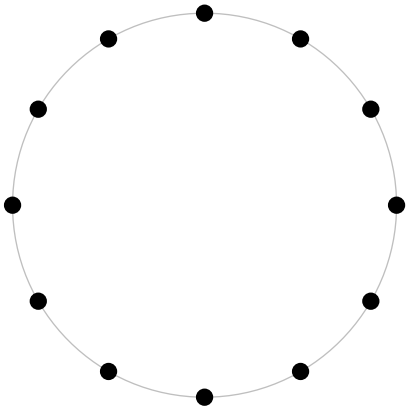
hop 1

pieces ☐



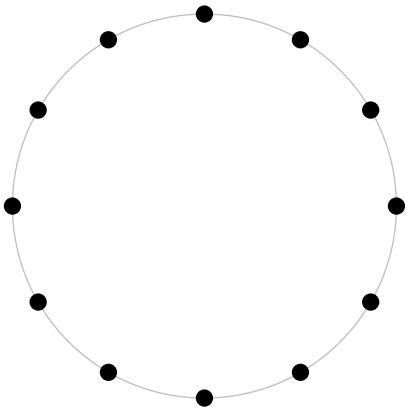
hop 2

pieces ☐



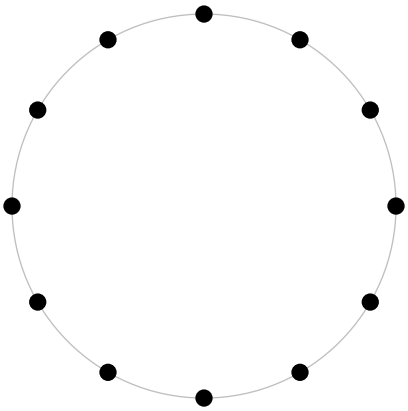
hop 3

pieces ☐



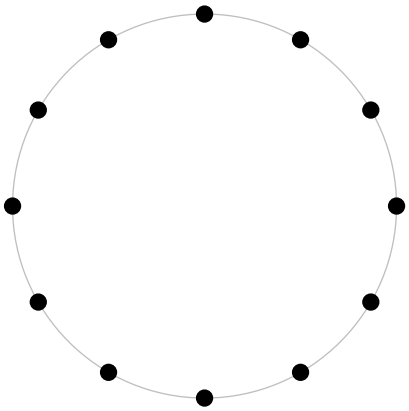
hop 4

pieces ☐



hop 5

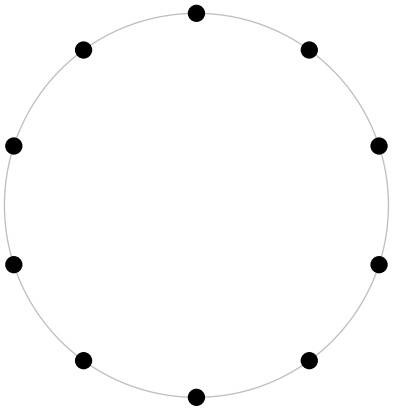
pieces ☐



hop 6

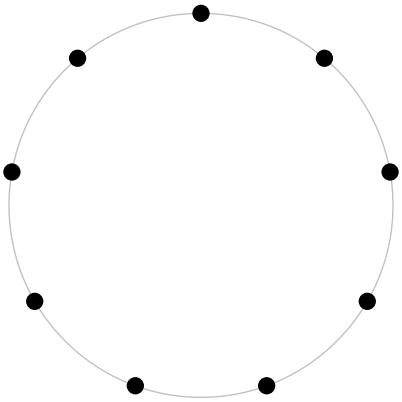
pieces ☐

Problem 2: Draw each of these, and write how many pieces it has.



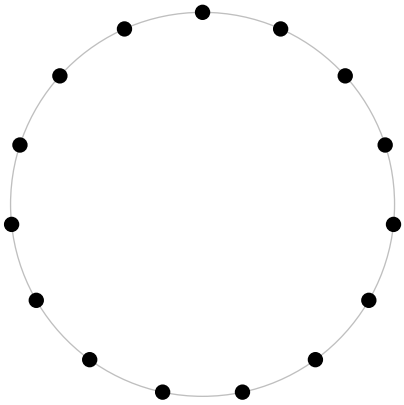
10 dots, hop 4

pieces ☐



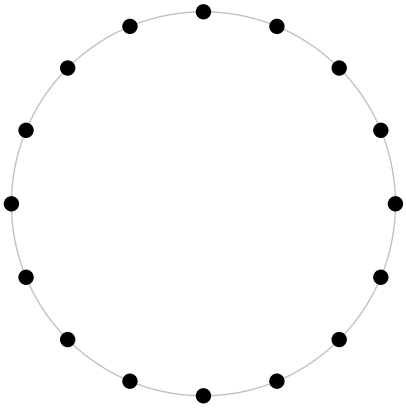
9 dots, hop 6

pieces ☐



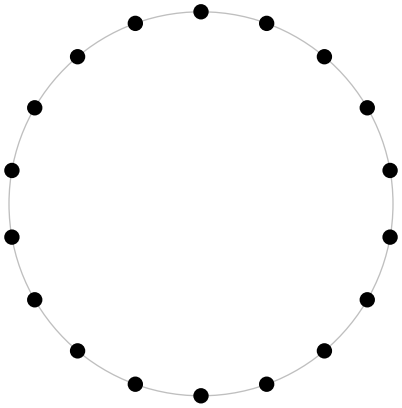
15 dots, hop 10

pieces ☐



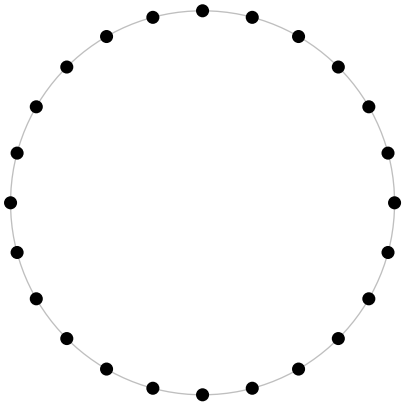
16 dots, hop 12

pieces ☐



18 dots, hop 8

pieces ☐



24 dots, hop 9

pieces ☐

Problem 3: Find how many pieces each of these drawings has, without drawing it. Then draw the first two on the rings to check.

20 dots, hop 8 pieces _____

24 dots, hop 10 pieces _____

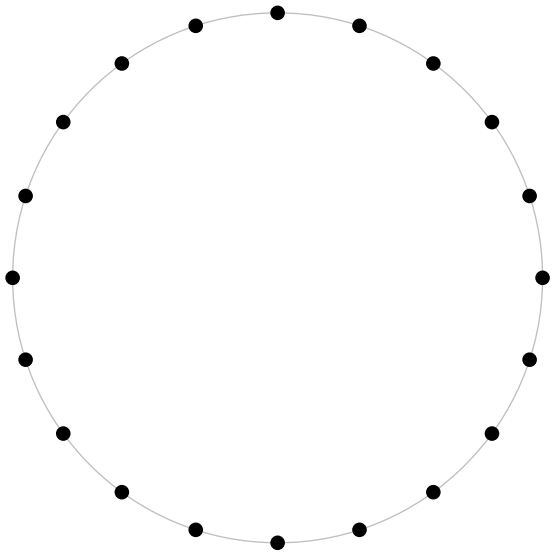
30 dots, hop 12 pieces _____

36 dots, hop 27 pieces _____

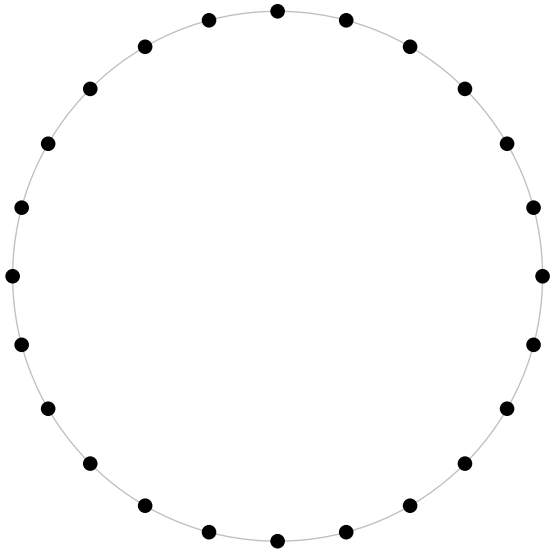
17 dots, hop 5 pieces _____

100 dots, hop 35 pieces _____

60 dots, hop 45 pieces _____



20 dots, hop 8



24 dots, hop 10

Problem 4: Cut out the two wheels on the last two pages and join them through the middle dots with a brass fastener. Setting the wheel to D means turning it until the inner A is next to the outer D. Each letter of a message on the inner wheel goes into code as the outer letter next to it, so with the wheel set to D, CAT goes into code as FDW. These codes were made with settings nobody wrote down. Find the messages, and find two different words that the last code could have come from.

R M A N F J B C J A F R C Q

__ __ __ __ __ __ __ __ __ __ __ __ __ __

C F N U E N Y X R W C B

__ __ __ __ __ __ __ __ __ __ __ __

O W E W W L S L F G G F T Q L Z W

__ __ __ __ __ __ __ __ __ __ __ __ __ __ __ __ __

T A Y L J W W

__ __ __ __ __ __ __

Y X I I L L K

__ __ __ __ __ __ __

M R O O B

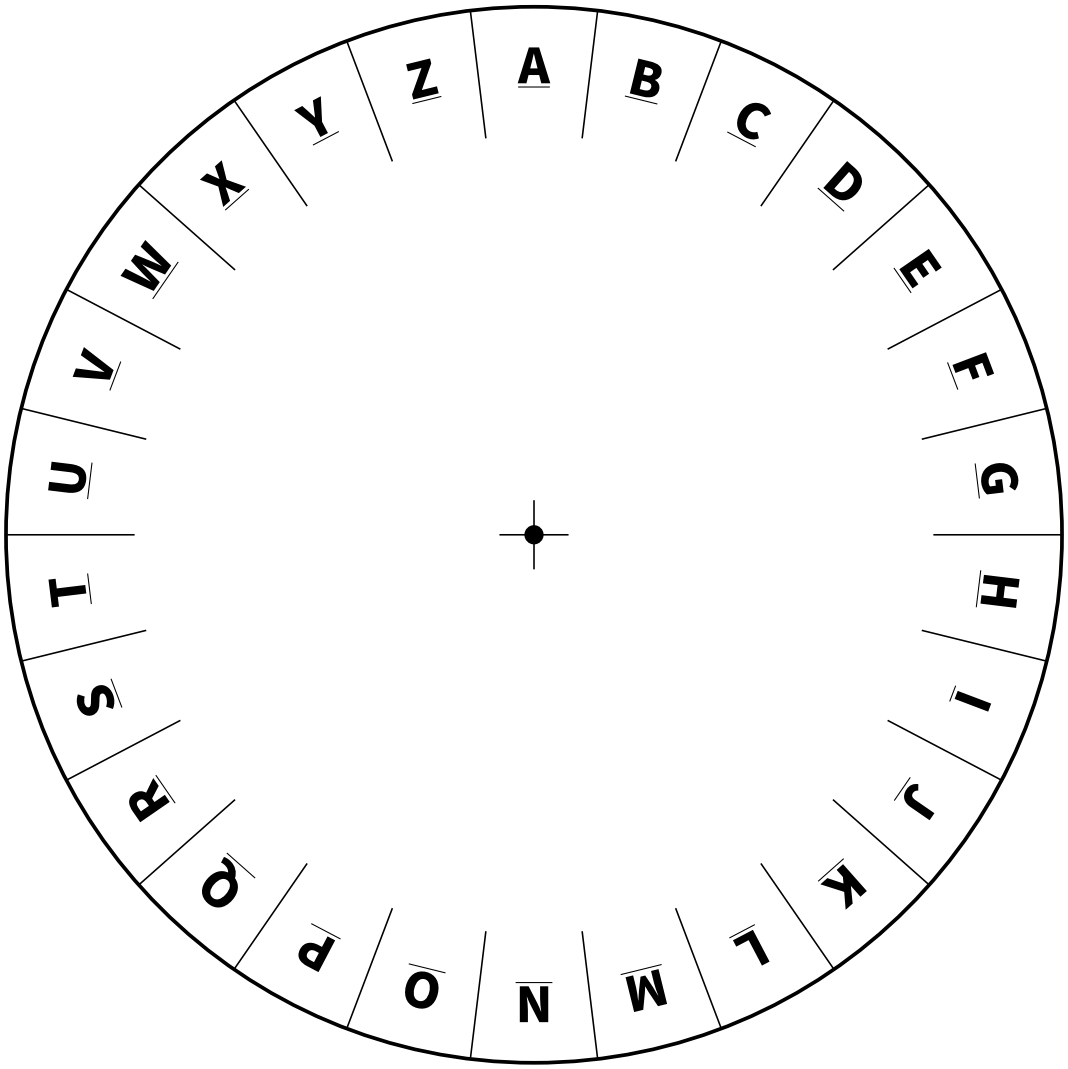
__ __ __ __ __

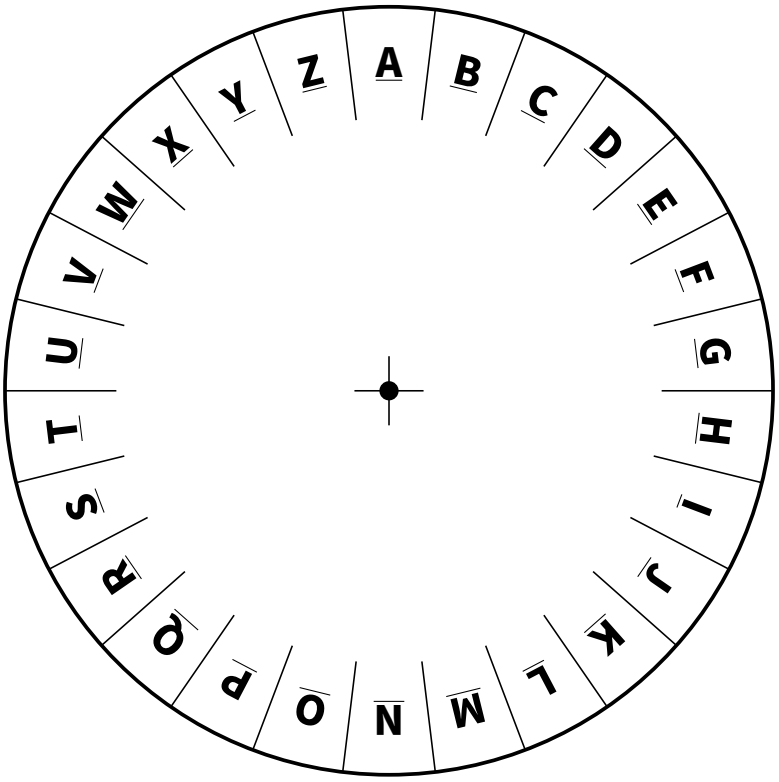
__ __ __ __ __

Problem 6: Put a word into code with the wheel set to H, and then put the result into code again with the wheel set to K. Which single setting gives the same result in one go? Find the single setting for each pair in the table, and the missing second setting in the last row.

first setting	second setting	single setting
H	K	
D	F	
T	M	
P	L	
H		A

Problem 7: Write a rule that gives the number of pieces from the number of dots and the hop. Explain why your rule always works.


A
B
C
D
E
F
G
H
I
J
K
L
M
N
O
P
Q
R
S
T
U
V
W
X
Y
Z


A
B
C
D
E
F
G
H
I
J
K
L
M
N
O
P
Q
R
S
T
U
V
W
X
Y
Z